Bryce Hackel

brycehackel.me | brycemhackel@gmail.com | linkedin.com/in/bryce-hackel | 408-931-4564

Education

University of California, San Diego — La Jolla, CA

M.S. Computer Science; GPA: 3.8 — *Sep. 2025 – Expected Mar. 2027*

B.S. Computer Engineering, Education Studies Minor; GPA: 3.92 — *Sep. 2021 – Jun. 2025*

Experience

Robotics Researcher — Jan. 2026 – Present

Prof. Michael Yip - Advanced Robotics and Controls Lab — *La Jolla, CA*

- **N2D Glove: Multi-Finger 2D Force-Feedback**: [Submitted IROS 2026] Co-designed the first multi-finger glove with per-finger 2D force feedback for robotic hand teleoperation. Added automatic joint zeroing to get accurate force-vector rendering. Debugged embedded firmware for six brushless DC motors with field-oriented control.
- **ROS Teleoperation Pipeline**: Built ROS-based force feedback pipeline to teleoperate a Franka arm with Inspire robotic hand. Increased haptic feedback rate from 10 Hz to 100 Hz.

Electrical Team Lead — Jul. 2025 – Present

Triton Droids at UC San Diego — *La Jolla, CA*

- **Force-Sensing Robotic Hand**: Leading a 5-engineer team building a $200 VR-teleoperated hand with per-fingertip Force-Sensing Resistors calibrated to 1 N for force-aware grasping.
- **Power Distribution Board PCB**: Leading design of a safe PDB to distribute 48 V up to 30 A across the robot, integrating a Jetson Orin with RobStride motors' CAN network.

Wireless Embedded Systems Engineer — Nov. 2024 – Nov. 2025

AquilX (Part Time) — *San Diego, CA*

- **Scalable System Test Automation**: Designed a system-level test platform to characterize BLE performance and amperometry of eight continuous glucose monitors (CGMs) in parallel. Wrote controller firmware in C for Zephyr RTOS on nRF52; built a Python/PyQt6 UI; coordinated outsourced PCB design.
- **Low-Power PWM DAC for Potentiostat**: Developed a configurable DAC using PWM and RC filtering to replace a fixed divider, increasing power by $< 5\,\%$ while maintaining 6-bit resolution.
- **BLE over UART**: Created a Python library to read/write/subscribe to BLE devices via an external nRF52 central over UART, avoiding issues with inconsistent laptop BLE controllers.
- **Standard Operating Procedures**: Authored SOPs documenting processes to meet regulatory requirements.

R&D Systems Engineering Intern — Jun. 2024 – Sep. 2024

Boston Scientific — *Santa Clarita, CA*

- **Sleep Detection Feature**: Created and pitched an accelerometer-based sleep-detection feature for next-gen spinal cord stimulators (SCS) to adjust stimulation dynamically.
- **Power-Efficient Machine Learning**: Built a data pipeline in C++ from accelerometer on-device to Python off-device, then optimized the RNN to reduce power by 98 % while maintaining $> 95\,\%$ accuracy.

Projects

- **Vision + Voice EMG Prosthetic Hand**: Battery-powered XIAO ESP32 Sense in the hand transcribes audio using Groq Cloud and sends transcript+image to Claude API for automatic grip selection using object recognition. EMG envelope threshold controls 8 Feetech SCS0009 servos in a modified AmazingHand to toggle the grip on/off.
- **Extreme Low-Power "AirTag" BLE Project**: Firmware for an STM32 with low-power I^2C accelerometer driver, interrupts, and deep-sleep modes. Achieved 70 µA static draw at 3 V (93 days).
- **Modded Open-Source iOS YouTube App**: Developed well-being oriented features in Swift and Obj-C: hidden tabs, in-player loop button, gesture controls, and optional UI hiding, to reduce distractions.

Skills

Primary Languages: Python, Embedded C/C++ | Swift, JavaScript

Technologies: ROS2, MuJoCo, SimpleFOC, BLE, CAN, JTAG/SWD, Altium, KiCAD, Fusion 360, LaTeX

Interests: China/Chinese language, Scaling manufacturing in the US, Humanoid robotics, Public transit